

Keypad Based Security Lock System

Internship Project

Intern Details

Field	Details
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Domain	Embedded Systems
Project Title	Keypad Based Security Lock System

Project Overview

This project implements a password-protected security lock using Arduino Uno, 4×4 Keypad, 16×2 LCD Display, and SG90 Servo Motor.

The user enters a password using the keypad. Arduino validates the password and controls the servo motor to lock or unlock the door.

Objectives

- Learn keypad interfacing
- Learn LCD interfacing
- Learn servo motor control
- Implement password authentication
- Develop an embedded security application

Components Required

1. Arduino Uno
2. 4×4 Matrix Keypad
3. 16×2 LCD Display
4. SG90 Servo Motor
5. 10K Potentiometer
6. Breadboard
7. Jumper Wires

Software Used

- Arduino IDE
 - Tinkercad
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Circuit Connections

Keypad Connections

Keypad Pin	Arduino Pin
R1	D9
R2	D8
R3	D7
R4	D6
C1	D5
C2	D4
C3	D3
C4	D2

Servo Motor Connections

Servo Pin	Arduino
VCC	5V
GND	GND
Signal	D10

LCD Connections

LCD Pin	Arduino
RS	D12
E	D11
D4	A0
D5	A1

LCD Pin	Arduino
D6	A2
D7	A3
VSS	GND
VDD	5V
RW	GND

Working Principle

1. User enters password using keypad.
2. Password is displayed as '*'.
3. User presses '#' to submit.
4. Arduino compares entered password with stored password.
5. If password is correct:
 6. LCD displays "Access Granted"
 7. Servo rotates and unlocks the door.
8. If password is wrong:
 9. LCD displays "Access Denied"
 10. Door remains locked.

Default Password

1234

Features

- Password Authentication
 - LCD User Interface
 - Servo Controlled Door Lock
 - Access Granted Message
 - Access Denied Message
 - Password Reset using '*'
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Project Output

Correct Password

Input: 1234#

Output: Access Granted

Wrong Password

Input: 1111#

Output: Access Denied

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